Najdi Frli

Technical report

Bilingual recycling guidance, walking and driving routes, JSP journeys, and access planning for Skopje

Design principle A model suggestion is not disposal permission. Routes require confirmed items and local policy; recycling and specialist destinations require explicit asset acceptance. Residual-waste bin navigation has an acknowledged acceptance exception.

Product	Citizen journeys, container check-ins, City planning, moderated dumping reports and optional activity accounts
Languages	English and Macedonian, with separate frozen text-suggestion heads
Travel	Walking, direct driving, parking-assisted driving and scheduled JSP buses
Architecture	Build-free browser client, one FastAPI service, SQLite and versioned offline artifacts
Evidence	Reproducible source snapshots, automated checks and explicit review limits

Prepared from the implementation and checked-in evidence on 17 October 2026. Source snapshots, experiment dates and historical validation records retain their original provenance. The GitHub repository URL remains unchanged.

Product and decision boundaries

Najdi Frli connects an item description to a suitable collection destination and helps City users compare access gaps. The resident flow covers glass, plastic/metal, paper/cartons, ordinary residual waste, electronics and wearable-clothes donation. A resident can select a known stream or request optional text/photo help, confirm item and condition, choose an origin, and plan up to six basket entries.

The serving order is local specialist and exact-item policy, optional broad model help, explicit user confirmation, then asset-level destination matching. Hazardous, held and unsupported descriptions block a trip. General-waste navigation requires an explicit warning and a recorded mixed-waste container. Electronics and clothes use specifically accepting facilities and condition-aware guidance. A broad label never establishes destination acceptance.

Current learned models

The served English and Macedonian models are the `v3-audit` frozen MiniLM six-class heads. Each language has its own encoder, logistic head and acceptance gate. Scores are temperature-scaled on development data, not established field probabilities. The gate can abstain; policy and manual selection remain usable when the models are disabled. Earlier v2 five-seed comparisons remain historical development evidence, not measurements of the current serving heads.

The current serving report measures the model fallback after policy decisions. Its denominators differ from closed-set accuracy or overall product accuracy:

Diagnostic set	Supported fallback	Accepted	Correct	Held false accepts
English development	30	18	18	0
Macedonian development	42	14	14	0
English manual	23	18	16	2
Macedonian manual	15	13	10	3

Development supported-fallback coverage is 60.0% in English and 33.3% in Macedonian. The Macedonian set contains correlated spelling variants: concept-weighted coverage is 32.5%. Manual accepted accuracy is 88.9% and 76.9%, respectively. Nonzero held-item accepts are a real limitation; downstream confirmation and policy checks remain required. These are development diagnostics with overlap and labeling limits, not independent field validation. See the exact serving report and the protocol and historical comparisons.

Language and photo assistance

English/Macedonian switching preserves the basket, origin and route. Machine IDs remain independent of translated presentation. Known Cyrillic and Latin Macedonian forms normalize at policy boundaries while preserving hazard and contamination cues; unknown prose is not silently translated. Optional photo analysis uses the configured provider and the same confirmation requirements. Invalid provider envelopes produce controlled failures; image preprocessing runs outside the async request loop.

Evidence limit Neither a model score nor a nearby marker proves that a container is operational or appropriate. Local policy review, native-language review and field access checks remain separate work.

Journeys and container conditions

All modes revalidate policy and remove closed or currently excluded assets before shortlisting. The common shortlist takes the three nearest compatible source sites per item type, deduplicated to at most 18 sites. This bounded heuristic does not establish a city-wide optimum. ORS supplies actual directed network routes. Missing keys, disconnected paths, invalid output or quota failures produce explicit errors, never invented walking or driving geometry.

Walking and driving

Direct trips minimize directed matrix travel time over covered-item/last-site states. Final directions supply displayed time and distance, which may differ from the matrix estimate because arrival-direction state is not retained between stops. Trips end at the final collection point. Gaps to recorded entrances are disclosed and excluded from totals, not drawn as verified paths.

Parking-assisted driving considers two eligible mapped parking points per candidate site within 800 m discovery distance. Actual walking connections must take at most ten minutes. Intermediate return walks to the car are included; the final return is excluded and the car's final location is shown. The 144-point public-parking snapshot derives from OSM with access and entrance filters. Space availability is always unknown; fees/hours are source tags, not live checks. A stale or invalid catalog disables this option without disabling direct driving. See parking limits and refresh instructions.

Scheduled JSP journeys

JSP's published GTFS timetable needs no key; ORS still requires a server-side key for walking. Each collection leg permits a direct bus or one same-stop transfer, bounded nearby-stop search and walking connections of at most 15 minutes. The greedy next-drop-off choice is not globally optimal. Times display in Europe/Skopje. Missing or invalid bus shapes produce a warning and textual itinerary.

These are scheduled journeys, not live arrivals. The planner trusts actual calendars and exceptions, rejects conflicting calendar records, and fails when data is stale or service is unavailable. The October 17 feed had service through October 27, despite a broader metadata claim. Real feed archives remain ignored runtime data because redistribution terms were not established. See source evidence and search limits.

Completion and shared status

Residents explicitly mark visits or drop-offs. The last drop-off ends the trip; Finish trip ends early without treating remaining items as disposed. Completion clears the route and offers an optional check-in for visited containers. No status is submitted by default and no container photo is required.

A full report immediately excludes that asset from everyone's new routes for up to 12 hours, labeled as unreviewed community information. Other assets at the site remain eligible. Reviewers can reject false reports; a newer approved available report can supersede an older full report. Expiry means unknown, not available. Existing routes show changes on status refresh and require explicit replanning. Dumping reports follow separate approval and never become route destinations. See expiry, moderation, retry and retention rules.

Trip scope Travel totals exclude parking search/payment and unverified access gaps. JSP elapsed time includes waits and three minutes per intermediate drop-off; walking and driving trips exclude disposal time.

City analysis and source evidence

City Mode estimates per-material walking access and compares hypothetical upgrades using prepared artifacts and a separately pinned pedestrian graph. It does not call the live routing provider. Community reports and citizen trips do not silently alter modeled coverage percentages.

Population-weighted walking coverage

Let p_i be modeled population at demand location i, and $d(i, S_m)$ the minimum directed walking time to a compatible site for material m, including terminal connectors. The metric is

$$\text{Coverage}_m(600) = 100 \frac{\sum_i p_i \, \mathbf{1}\{d(i, S_m) \le 600\}}{\sum_i p_i}.$$

The graph splits at demand/site fractions. Reverse multi-source Dijkstra preserves direction and counts each location once. Walking speed is 4.8 km/h. Unverified connectors up to 100 m are terminal costs, not shortcuts. The denominator includes about 522,421 modeled residents across ten municipalities, including about 8,620 disconnected residents. NoData areas remain unknown.

Material stream	Current coverage	Five hypothetical upgrades
Glass packaging	58.14%	59.34%
Plastic bottles / metal cans	34.94%	36.88%
Paper / cardboard / Tetra Pak	5.04%	11.44%
General waste	Unavailable	Unavailable

General-waste estimates are unavailable because a positive reviewed acceptance profile is missing. Citizen navigation to a recorded residual-waste bin does not supply that missing planning evidence.

Upgrade scenarios

Budgets from zero to twenty add one reviewed stream to eligible existing general-waste sites hypothetically. Overlapping demand IDs are unioned before population is summed. The objective seeks a 50% municipal floor where feasible, otherwise compares sorted municipal coverage lexicographically, then total covered population and stable IDs. Small spaces are enumerated exactly; larger searches use deterministic heuristics and disclose optimality. The floor remains infeasible under the present model. Scenarios do not change source acceptance or authorize installations.

Provenance and refresh

The City snapshot contains 1,596 source assets grouped into 1,583 sites. Citizen discovery adds separately documented electronics and clothes destinations. Source profiles, operation status and reuse-review flags stay distinct. Exact-byte manifests and input/code hashes detect incompatible analytical artifacts. Renaming the app does not change source acquisition dates or experiment records.

Population derives from WorldPop Global2 R2025A v1 constrained alpha, University of Southampton (DOI:10.5258/SOTON/WP00839; CC BY 4.0). The analytical network uses OpenStreetMap contributors via the September 18, 2026 Geofabrik extract (ODbL 1.0); parking uses the October 16 extract. Source-specific reuse limits remain in force. See coverage, upgrades, geography and data contracts.

Architecture, privacy and verification

The build-free client talks to one FastAPI service. Policy, frozen heads, source snapshots and compact City reports load at startup. Training and graph construction run offline. SQLite stores community reports, quotas and optional activity accounts. The deployment assumes one application instance; multiple replicas need shared storage and coordinated limits.

Private and public data

Trip origins, baskets and route geometry remain in page memory. Planning sends coordinates to ORS through the disclosed action; descriptions are not sent to that provider. Sensitive API responses, including validation failures and early rejections, carry `Cache-Control: no-store`. Account mutations use an explicit action header and same-site protections. The renamed client uses `X-Najdi-Frli-Action`; the previous header remains accepted for compatibility. Existing browser language preferences migrate to the renamed storage key.

Approved dumping locations are intentionally public; photos remain reviewer-only. Reports/photos expire after 30 days on database access. Container exclusions expire sooner according to condition. Optional accounts save material/date activity, not route history. Paid requests have persistent client/global quotas, bounded timeouts and concurrency limits. These do not establish production availability against distributed abuse. See accounts, community data and runtime boundaries.

October maintenance audit

The app, package metadata and documentation now use Najdi Frli. The repository URL, internal Python import path and established `SRA_` configuration names remain stable. Documentation separates current behavior from historical measurements and identifies the served v3 audit models. The code audit fixes sensitive-response cache headers, invalid timetable records, malformed photo-provider responses, blocking image preprocessing, reviewer action races, stale map results and duplicate report retries. Focused regressions cover these changes.

Reproducible checks

CI runs lock verification, lint, formatting, backend and JavaScript tests, infrastructure/geography validation, exact coverage/upgrade reproduction, Chromium tests and Docker startup smoke with both learned heads required. The preceding merged revision `c626c8d` passed 737 backend, 429 browser and 21 Node tests plus Docker smoke; those counts belong to that revision. This audit's results are recorded separately in QUALITY and CI status, with historical evidence clearly labeled.

Use `uv sync --all-groups --locked`, then `uv run python scripts/run_local.py`. Run `make check` with Node 22 and Make, or PowerShell equivalents in README. Enable `SRA_BROWSER_TESTS=1` for browser tests. Reproduce City artifacts using `scripts/prepare_coverage.py --check` and `scripts/prepare_upgrades.py --check` with the `geo` group. Provider smoke scripts are opt-in and consume quota.

Remaining evidence gaps

Provider and map fixtures verify application behavior, not physical access or live service quality. Live probes establish individual integrations, not completed journeys. Native Macedonian review, independent local model evaluation, container/entrance checks, physical-phone testing and hosted HTTPS rehearsal remain outstanding. Assess policy, model abstention, destination acceptance and actual access separately.

Current claim Najdi Frli is a reproducible prototype with explicit uncertainty, bounded routing and tested integration behavior. Field operation, independent language quality and authoritative disposal guidance still need external evidence.

Entry points: contributor guide, status, API, ML evidence, validation.